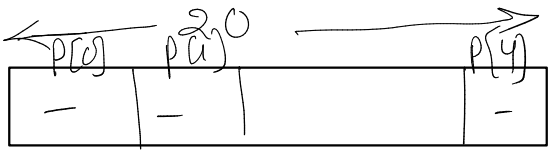
```
int *p;
p = malloc(20);
//p = malloc(5 * sizeof(int));
```

(2 * 10)



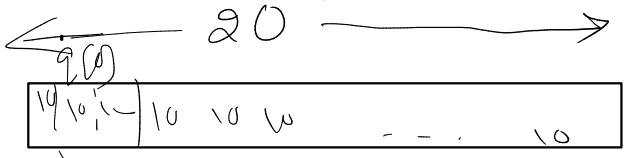
5500

p
5500

```
main(){
int *q;
q = calloc(1,20);
// q = calloc(5, sizeof(int));
```

10, 2

2, 10



6000

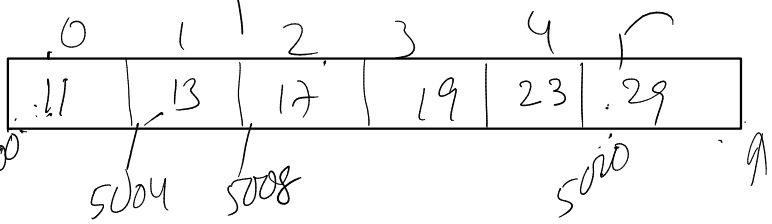
q

malloc, calloc	New Alloc
free	Unreserve/Deallocate
realloc	Resize

malloc VS calloc

```
*arr[i] = data;
arr++;
```

arr



int *temp

5000 5004

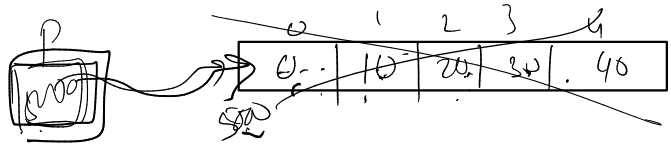
5000	24
6080	36
.	.
.	.
.	.

void *malloc

void *realloc(. .)

System call

```
int *p = malloc(20);
for(i=0; i<5; i++) p[i] = i*10;
```

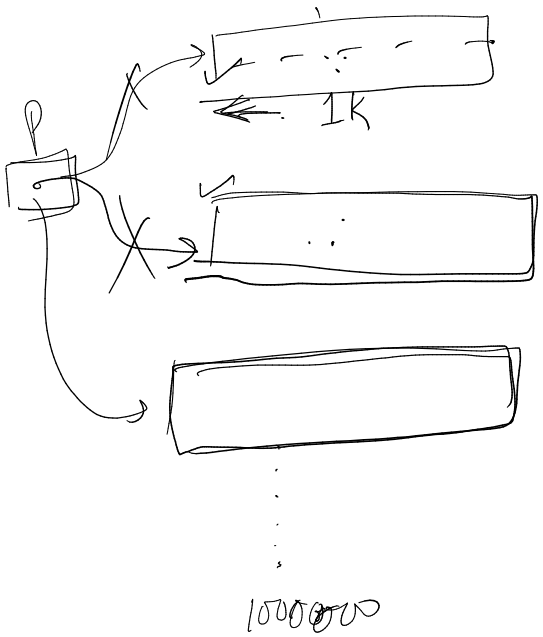



```
pf("%d %d %d", p[0], p[1], p[2]); // 0 10 20
free(p);
```

Dangling Pointer

```
pf("%d %d %d", p[0], p[1], p[2]); // 0 10 20 ✓
```

```
p = malloc(n);
// free(p);
```



Memory leak

---

n=20

```
char *p = malloc(n);
strcpy(p, "VECTOR");
n = strlen(p);
p = realloc(p, n+1);
free(p+n+1); // seg fault
```



5000	20
6000	20
8500	40

n=0

p = realloc(p, n); // free(p)

char *p = NULL;

p = realloc(p, 20); // malloc(20);

realloc(NULL, Not zero) // malloc. ✓

realloc(Not NULL, 0); // free.

realloc(NULL, Non zero); // realloc / resize

---

char str[20];

sf("%s", str) vs gets(str) vs fgets(str, n, stdin)

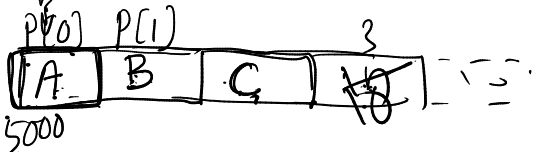
char *p;

sf("%s", p) gets(p) , fgets(p, -, -)

---

p = NULL

5000



i = 01234

ABC\0

(i) 2D Array

(ii)